Class (Section): _______________ **Date: ______________**
Name: _______________________ **Deep Learning**
Roll No: _____________________ **Dr. Zohair Ahmed**

Activity Sheet 1: Backpropagation

We have this network:

- Inputs: x_1 = 1 ; x_2 = 0
- Target (label): y = 1

Hidden neuron 1:	**Hidden neuron 2:**	**Output neuron:**
Weights: w_{11} = 0.1, w_{12} = 0.2 Bias: b_1= 0.1	Weights: w_{21} = 0.3 w_{22}= 0.4 Bias: b_2 = 0.1	Weights: v_1 = 0.5 v_2 = 0.6 Bias: b_3 = 0.2

- **Activation function (for all neurons):** $\sigma(z) = \frac{1}{1+e^{-z}}$
- **Loss function:** $L = \frac{1}{2}(y - \hat{y})^2$

For this worksheet we **assume the forward pass has already been done** and the following values are given:

- $z_1 = 0.2, \quad h_1 = 0.5498$
- $z_2 = 0.4, \quad h_2 = 0.5987$
- $z_3 = 0.8341, \quad \hat{y} = 0.6973$
- $L \approx 0.0458$

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1. From Loss L to Prediction $\hat{y}$
Step 1.1 – Write the loss function in terms of $\hat{y}$:

$$L(\hat{y}) = \frac{1}{2}(y - \hat{y})^2$$

a) Substitute y = 1 and $\hat{y}$**:** = 0.6973 (just to check the value):

$$L = \frac{1}{2}(1 - 0.6973)^2 = \frac{1}{2}(\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_)^2$$

$L \approx$ ____________________(should be around 0.0458)

Step 1.2 – Derivative of $\boldsymbol{L}\ w.r.t.\boldsymbol{\hat{y}}$

$$u = y - \hat{y}$$
$$L = \frac{1}{2}u^2$$

(a) $\frac{dL}{du} = \frac{d}{du}\left(\frac{1}{2}u^2\right)$; $\frac{dL}{du} = u$

(b) $\frac{du}{d\hat{y}} = \frac{d}{d\hat{y}}(y - \hat{y})$
Since y is constant with respect to $\hat{y}$,

$$\frac{du}{d\hat{y}} = -1$$

a) $\boxed{u}$
b) $\boxed{-1}$

1. Apply the chain rule:

$$\frac{\partial L}{\partial \hat{y}} = \frac{\mathrm{d}L}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}\hat{y}} = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$$

3. Simplify the result:

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y$$

b) Now plug in the numbers: $\hat{y}$ = 0.6973,; y = 1

$$\frac{\partial L}{\partial \hat{y}} = 0.6973 - 1 = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$$

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2. From $\hat{y}$ to Output Pre-activation z_3
We know: $\hat{y} = \sigma(z_3)$.
Step 2.1 – Sigmoid derivative

Write the derivative formula:

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1-\sigma(z))$$

So,

$$\frac{\partial \hat{y}}{\partial z_3} = \hat{y}(1-\hat{y})$$

a) Compute 1 - $\hat{y}$:

$$1-\hat{y} = 1-0.6973 = \_\_\_\_\_\_\_\_\_\_\_\_$$

b) Multiply:

$$\frac{\partial \hat{y}}{\partial z_3} = 0.6973 \times (\_\_\_\_\_\_\_\_\_\_\_\_) = \_\_\_\_\_\_\_\_\_\_\_\_$$

Step 2.2 – Error signal at the output: $\delta_3 = \frac{\partial L}{\partial z_3}$

δ3 measures **how much the output z3 contributed to the loss**.

Use chain rule:

$$\frac{\partial L}{\partial z_3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_3}$$

c) Substitute your results:

$$\delta_3 = \frac{\partial L}{\partial z_3} = (\_\_\_\_\_\_) \times (\_\_\_\_\_\_) = \_\_\_\_\_\_$$

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3. From z_3 to Hidden Activations h_1, h_2

Recall:

$$z_3 = v_1 h_1 + v_2 h_2 + b_3$$

3.1 Compute $\frac{\partial L}{\partial h_1}$

1. Local derivative:

$$\frac{\partial z_3}{\partial h_1} = v_1 = \_\_\_\_\_\_\_\_\_\_\_\_$$

2. Chain rule:

$$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_1}$$

a) Substitute:

$$\frac{\partial L}{\partial h_1} = (\_\_\_\_\_\_\_) \times (\_\_\_\_\_\_\_) = \_\_\_\_\_\_\_$$

3.2 Compute $\frac{\partial L}{\partial h_2}$

1. Local derivative:

$$\frac{\partial z_3}{\partial h_2} = v_2 = \_\_\_\_\_\_\_$$

2. Chain rule:

$$\frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial z_3} \cdot \frac{\partial z_3}{\partial h_2}$$

b) Substitute:

$$\frac{\partial L}{\partial h_2} = (\_\_\_\_\_\_\_) \times (\_\_\_\_\_\_\_) = \_\_\_\_\_\_\_$$

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4. From h_1, h_2 to z_1, z_2

We have:

- $h_1 = \sigma(z_1)$
- $h_2 = \sigma(z_2)$

4.1 Compute $\frac{\partial h_1}{\partial z_1}$ **and** $\delta_1 = \frac{\partial L}{\partial z_1}$.

1. Sigmoid derivative:

$$\frac{\partial h_1}{\partial z_1} = h_1(1 - h_1)$$

a) Compute 1 - h_1:

$$1 - h_1 = 1 - 0.5498 = \_\_\_\_\_\_\_$$

b) Multiply:

$$\frac{\partial h_1}{\partial z_1} = 0.5498 \times (\_\_\_\_\_\_\_) = \_\_\_\_\_\_\_$$

2. Chain rule for δ_1:

$$\delta_1 = \frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial h_1} \cdot \frac{\partial h_1}{\partial z_1}$$

c) Substitute:

$$\delta_1 = (\_\_\_\_\_\_\_) \times \_\_\_\_\_\_\_) = \_\_\_\_\_\_\_$$

4.2 Compute $\frac{\partial h_2}{\partial z_2}$ and $\delta_2 = \frac{\partial L}{\partial z_2}$.

1. Sigmoid derivative:

$$\frac{\partial h_2}{\partial z_2} = h_2(1 - h_2)$$

a) Compute 1 - h_2:

$$1 - h_2 = 1 - 0.5987 = \_\_\_\_\_\_\_$$

b) Multiply:

$$\frac{\partial h_2}{\partial z_2} = 0.5987 \times (\_\_\_\_\_\_\_) = \_\_\_\_\_\_\_$$

2. Chain rule for δ_2:

$$\delta_2 = \frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial h_2} \cdot \frac{\partial h_2}{\partial z_2}$$

c) Substitute:

$$\delta_2 = (\_\_\_\_\_\_\_) \times (\_\_\_\_\_\_\_) = \_\_\_\_\_\_\_$$

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5. Gradients for Output Layer Parameters $\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{b_3}$

Recall:

$$z_3 = v_1 h_1 + v_2 h_2 + b_3$$

So:

$$\frac{\partial z_3}{\partial v_1} = h_1, \frac{\partial z_3}{\partial v_2} = h_2, \frac{\partial z_3}{\partial b_3} = 1$$

Using $\frac{\partial L}{\partial v_i} = \frac{\partial L}{\partial z_3} \cdot \frac{\partial z_3}{\partial v_i}$

Step 5.1 : $\boldsymbol{\frac{\partial L}{\partial v_1}}$

$$\frac{\partial L}{\partial v_1} = \delta_3 \cdot h_1 = (\_\_\_\_\_\_\_\_\_\_\_) \times 0.5498 = \_\_\_\_\_\_\_\_\_\_\_$$

Step 5.2 : $\boldsymbol{\frac{\partial L}{\partial v_2}}$

$$\frac{\partial L}{\partial v_2} = \delta_3 \cdot h_2 = (\_\_\_\_\_\_\_\_\_\_\_) \times 0.5987 = \_\_\_\_\_\_\_\_\_\_\_$$

Step 5.2 : $\boldsymbol{\frac{\partial L}{\partial b_3}}$

$$\frac{\partial L}{\partial b_3} = \delta_3 \cdot 1 = \_\_\_\_\_\_\_\_\_\_\_$$

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6. Gradients for Hidden Layer 1 Parameters $\boldsymbol{w_{11}}, \boldsymbol{w_{12}}, \boldsymbol{b_1}$

Recall:

$$z_1 = w_{11}x_1 + w_{12}x_2 + b_1$$

So:

$$\frac{\partial z_1}{\partial w_{11}} = x_1, \frac{\partial z_1}{\partial w_{12}} = x_2, \frac{\partial z_1}{\partial b_1} = 1$$

And:

$$\frac{\partial L}{\partial w_{1j}} = \delta_1 \cdot x_j, \frac{\partial L}{\partial b_1} = \delta_1$$

Step 6.1 : $\frac{\partial L}{\partial w_{11}}$

$$\frac{\partial L}{\partial w_{11}} = \delta_1 \cdot x_1 = (\_\_\_\_\_\_\_\_\_\_\_\_) \times 1 = \_\_\_\_\_\_\_\_\_\_\_\_$$

Step 6.2 : $\frac{\partial L}{\partial w_{12}}$

$$\frac{\partial L}{\partial w_{12}} = \delta_1 \cdot x_2 = (\_\_\_\_\_\_\_\_\_\_\_\_) \times 0 = \_\_\_\_\_\_\_\_\_\_\_\_$$

Step 6.3 : $\frac{\partial L}{\partial w_{13}}$

$$\frac{\partial L}{\partial b_1} = \delta_1 \cdot 1 = \_\_\_\_\_\_\_\_\_\_\_\_$$

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7. Gradients for Hidden Layer 2 Parameters $\boldsymbol{w_{21}, w_{22}, b_2}$

Recall:

$$z_2 = w_{21}x_1 + w_{22}x_2 + b_2$$

So:

$$\frac{\partial z_2}{\partial w_{21}} = x_1, \frac{\partial z_2}{\partial w_{22}} = x_2, \frac{\partial z_2}{\partial b_2} = 1$$

And:

$$\frac{\partial L}{\partial w_{2j}} = \delta_2 \cdot x_j, \frac{\partial L}{\partial b_2} = \delta_2$$

Step 7.1 : $\frac{\partial L}{\partial w_{21}}$

$$\frac{\partial L}{\partial w_{21}} = \delta_2 \cdot x_1 = (\_\_\_\_\_\_\_\_\_\_\_\_) \times 1 = \_\_\_\_\_\_\_\_\_\_\_\_$$

Step 7.2 : $\frac{\partial L}{\partial w_{22}}$

$$\frac{\partial L}{\partial w_{22}} = \delta_2 \cdot x_2 = (\_\_\_\_\_\_\_\_\_\_\_\_) \times 0 = \_\_\_\_\_\_\_\_\_\_\_\_$$

Step 7.3 – $\frac{\partial L}{\partial b_2}$

$$\frac{\partial L}{\partial b_2} = \delta_2 \cdot 1 = \_\_\_\_\_\_\_\_\_\_\_\_$$

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8. Summary Table (Student Fills In)

Fill in the final values you computed:

Parameter	$\boldsymbol{\frac{\partial L}{\partial \text{parameter}}}$
v_1	
v_2	
b_3	
w_{11}	
w_{12}	
b_1	
w_{21}	
w_{22}	
b_2	